



DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

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CLASS

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INDEX
NUMBER

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PRELIMINARY EXAMINATION 2025 SECONDARY 4 EXPRESS

ADDITIONAL MATHEMATICS

4049/01

Paper 1

20 August 2025

2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Calculator Model

Answer **all** questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total marks for this paper is 90.

Total Marks

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formula for ΔABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2}bc \sin A$$

- 1 The line $y = x - 4$ and the curve $y = \frac{2}{x-3}$ intersect at the points A and B .
Find the length of AB .

[6]

2 (a) State the range of $\tan^{-1} p$. [1]

(b) Given that $\cos^{-1} k = x$, where $0 \leq x \leq \frac{\pi}{2}$, find the principal value of $\cos^{-1}(-k)$. [1]

3 Explain why $3x^2 + 15x + 20$ cannot be smaller than 1. [3]

- 4 **(a)** Express $\frac{x^2 + 3x + 2}{x^2(2x + 1)}$ in partial fractions. [5]

- (b)** Hence integrate $\frac{x^2 + 3x + 2}{x^2(2x + 1)}$ with respect to x . [3]

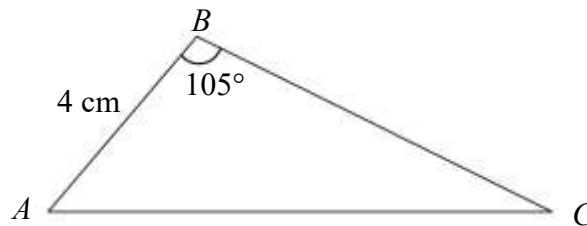
5 Solve the equation $3\log_5 y - \log_y 5 = 2$.

[4]

- 6 (a) Show that $\sin 105^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$. [3]

- (b) The diagram below shows triangle ABC , such that $AB = 4$ cm, angle $ABC = 105^\circ$ and the area of triangle ABC is $10 + 2\sqrt{3}$ cm².

Find BC in the form $\sqrt{2}(a\sqrt{3} + b)$, where a and b are integers. [3]



- 7 **(a)** Given that $u = \left(\frac{3}{2}\right)^x$, express $9^x + 6^x = 6(4^x)$ as a quadratic equation in u . [3]

- (b)** Hence, or otherwise, solve $9^x + 6^x = 6(4^x)$. [3]

- 8 The expression $2x^3 + ax^2 + bx + 10$, where a and b are constants, has a factor of $x + 2$ and leaves a remainder of 10 when divided by $x - 3$.
Find the values of a and of b . [4]

- 9 A cyclist starts from rest from a point A and travels in a straight line until she comes to a rest at a point B . During the motion, her velocity, v m/s, is given by $v = 12t - \frac{3}{2}t^2$, where t is the time in seconds after leaving A .

(a) Find the time taken for the cyclist to travel from A to B . [1]

(b) Find the distance AB . [3]

(c) Find the acceleration of the cyclist when $t = 5$. [2]

- 10** The equation of a curve is $y = xe^{-3x}$.

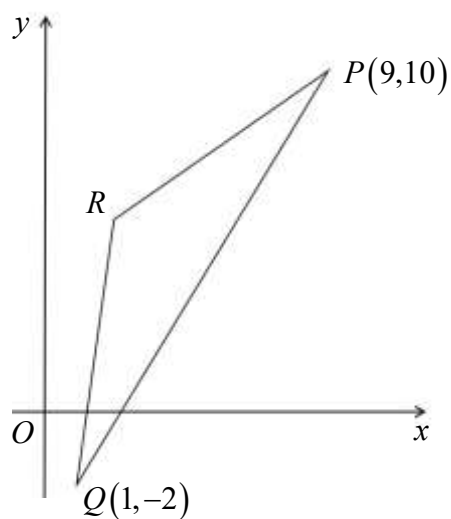
Find the stationary point(s) and determine the nature of the stationary point(s).

[6]

- 11 (a) Sketch the graphs of $y = -2\cos 3x + 1$ and $y = -\frac{2}{\pi}x + 1$ on the same axes, for $0 \leq x \leq \pi$. [5]

- (b) Explain how the number of solutions for $2\cos 3x = \frac{2}{\pi}x$ can be found using the graphs. [1]

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The diagram shows a triangle PQR with vertices $P(9,10)$ and $Q(1,-2)$. The point R lies on the perpendicular bisector of PQ and the equation of the line QR is $y = 8x - 10$.

(a) Find the equation of the perpendicular bisector of PQ . [4]

(b) Find the coordinates of R . [2]

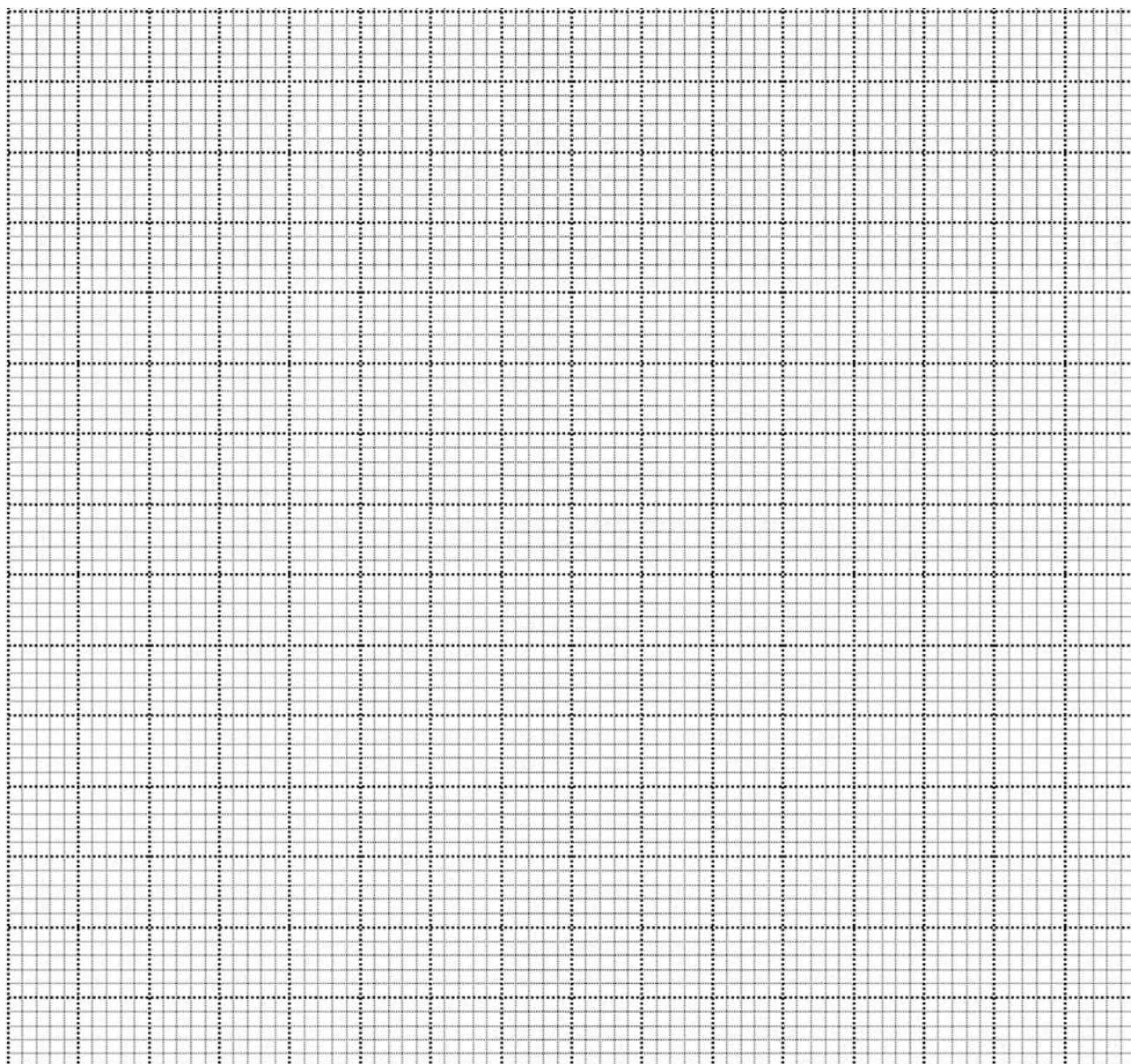
(c) Find the area of triangle PQR . [2]

- 13** The mass, m mg, of a radioactive substance decreases with time, t hours.
 It is known that m and t are related by the equation $m = Me^{-kt}$, where M and k are constants.
 The table below shows measured values of m and t .

t (hours)	1	2	3	4	5
m (mg)	6.55	5.36	4.40	3.60	2.94

- (a)** Plot $\ln m$ against t and draw a straight line graph.

[2]



Use your graph to estimate

(b) the initial mass of the substance,

[2]

(c) the value of k .

[2]

- 14** A curve is such that $\frac{d^2y}{dx^2} = -4\sin(2x + \pi)$ and the point $P\left(\frac{\pi}{2}, \pi\right)$ lies on the curve.

The gradient of the curve at P is 3.

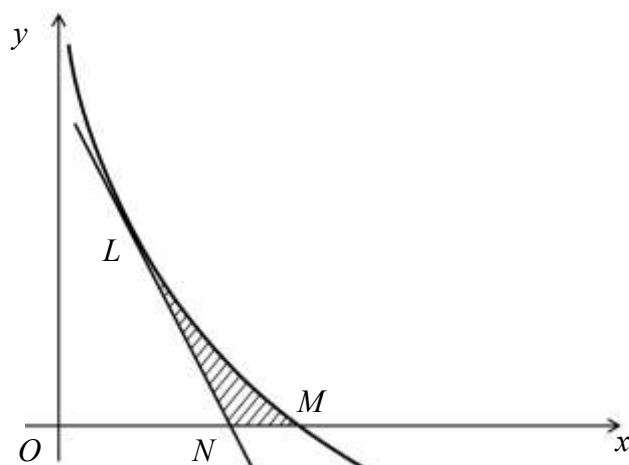
- (a)** Find the equation of the curve.

[5]

- (b)** Justify whether y is increasing when $x = 3$.

[2]

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The diagram shows part of the curve $y = \frac{16}{(x+1)^2} - 1$, cutting the x -axis at M .

The tangent at the point L on the curve cuts the x -axis at N .

The gradient of this tangent is -4 .

Calculate the area of the shaded region LMN .

[12]

Continuation of working space for Question 15.

END OF PAPER